DSC 140A - Homework 06

Homework 06 Instructions. You may collaborate with whomever, including AI. Each student must submit their own homework even if they worked with others. We will not be grading for correctness, but we will be grading for completeness. Please turn out a single PDF file that contains your solutions to all of the problems. We recommend using LaTeXwith our template. You can use markdown but make sure each problem is clearly labeled. For coding problems, we recommend using Jupyter notebooks, covert them into PDF, and then merge the PDF with your math problems solutions generated in LaTeX.

Problem 1.

In this problem, you will demonstrate that kernel ridge regression is equivalent to ridge regression performed in feature space by showing that they make the same predictions on a concrete data set.

We'll use the toy data set:

i	$\vec{x}^{(i)}$	y_i
1	$(0,2)^T$	1
2	$(1,0)^T$	1
3	$(0,-2)^T$	1
4	$(-1,0)^T$	1
5	$(0,0)^T$	-1

We will define

$$\vec{\phi}(\vec{x}) = (1, x_1^2, x_2^2, \sqrt{2}\,x_1, \sqrt{2}\,x_2, \sqrt{2}\,x_1x_2)^T.$$

It turns out that $\kappa(\vec{x}, \vec{x}') = (1 + \vec{x} \cdot \vec{x}')^2$ is a kernel for $\vec{\phi}$.

You can write code to perform any and all calculations in this problem. If you do, please show your code.

a) Learn a prediction rule $H_1(\vec{x})$ by performing ridge regression in the 6-dimensional feature space and report the optimal parameter vector $\vec{w}$. Use a regularization parameter of $\lambda = 2$.

b) Given a new point, $\vec{x} = (1,1)^T$, what is the prediction made by your ridge regressor? That is, what is $H_1(\vec{x})$? Show your calculations / code.

c) Compute the kernel matrix, K, for this data.

d) Learn a prediction function $H_2(\vec{x})$ by solving the kernel ridge regression dual problem; that is, by finding the optimal vector $\vec{\alpha}$. Recall that there is an exact solution: $\vec{\alpha} = (K + n\lambda I)^{-1}\vec{y}$. Report the $\vec{\alpha}$ that you find.

Note: the lecture slides had originally stated the solution without the n in $n\lambda I$. This was a typo and has been fixed.

e) Let $\vec{x} = (1,1)^T$, as before. What does your kernel ridge regressor predict for this point? That is, what is $H_2(\vec{x})$?

Hint: $H_2(\vec{x})$ should be the same as $H_1(\vec{x})$.

Problem 2.

Let X be a continuous random variable, and let Y be a binary random variable. Suppose the class conditional densities are know to be:

$$p_X(x \mid Y=0) = \begin{cases} 1/5, & \text{if } 0 \le x \le 2 \\ 1/3, & \text{if } 2 < x \le 3 \\ 1/15, & \text{if } 3 < x \le 7 \end{cases}$$

$$p_X(x \mid Y=1) = \begin{cases} 1/6, & \text{if } 0 \le x \le 1 \\ 1/8, & \text{if } 1 < x \le 5 \\ 1/6, & \text{if } 5 < x \le 7 \end{cases}$$

Suppose also that $\mathbb{P}(Y=0) = 0.4$ and $\mathbb{P}(Y=1) = 0.6$.

For what values of $x \in [0,7]$ will the Bayes classifier predict $y=1$?

Problem 3.

Let X be a continuous random variable, and let Y be a random class label (1 or 0). Recall that the Gaussian probability density function (pdf) is given by:

$$\mathcal{N}(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Suppose $p_X(x \mid Y=1)$ is a Gaussian pdf with $\mu = 2$ and $\sigma = 3$ and that $p_X(x \mid Y=0)$ is also Gaussian with $\mu = 5$ and $\sigma = 3$. Suppose, too, that $\mathbb{P}(Y=1) = \mathbb{P}(Y=0) = \frac{1}{2}$.

Recall that the Bayes error is the probability that the Bayes classifier makes an incorrect prediction. What is the Bayes error for this distribution? Show your work.

Hint: you'll want some way to compute the area under a Gaussian. You can use the tables that appear in the back of your statistics book, or you can use something like `scipy.stats.norm.cdf`. We'll let you read the documentation to see how to use it, but it may be helpful to remember that if F is the cumulative density function for a distribution with density f, then $\int_a^b f(x)\,dx = F(b) - F(a)$.

Problem 4.

The file linked below contains a data set of 150 samples. The first column contains a single continuous feature, X, assumed to have been drawn from an unknown probability density. The second column contains the binary class label Y.

`https://f000.backblazeb2.com/file/jeldridge-data/011-univariate_density_estimation/data.csv`

In this problem, use a histogram estimator with bins $[0, 1.5), [1.5, 3), \ldots, [13.5, 15)$ to estimate the requested probabilities. In each part, show your code and provide your reasoning.

Hint: you may find `np.histogram` useful.

a) Estimate $\mathbb{P}(Y=1 \mid X=6.271)$ directly.

b) Estimate $\mathbb{P}(Y=1 \mid X=6.271)$ by estimating all of: 1) the marginal density $p_X(x)$, 2) the class-conditional density $p_X(x \mid Y=1)$, and 3) the class prior $\mathbb{P}(Y=1)$, and then applying Bayes' rule.

c) For what values of $x \in [0, 15]$ will the Bayes classifier predict $y=1$?